

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the curve $y = \sqrt{x+1}$ on the interval $[0, 3]$. Set up integrals with respect to x and y which represent the arc length of the curve on $[0, 3]$.

2. Kyle attempts to find the arc length of $y = \sqrt{x+1}$ on $[-1, 3]$. He sets up the integral $L = \int_{-1}^3 \sqrt{\frac{5+4x}{4+4x}} dx$, but his professor tells him that there is an issue with his integral and that his integral may not actually represent the length of the curve. What is the problem with Kyle's work?

3. Consider the curve $y = \sqrt{x+1}$ on the interval $[0, 3]$.
- a) Suppose this curve was rotated around the x -axis and the y -axis. Draw pictures of the solids that are created by each of these rotations.
- b) Consider the case where the curve has been rotated about the x -axis. Setup, but do not evaluate, an integral with respect to x and an integral with respect to y that represent the surface area of the figure.
- c) Consider the case where the curve has been rotated about the y -axis. Setup, but do not evaluate, an integral with respect to x and an integral with respect to y that represent the surface area of the figure.

4. Consider the curve C given by $y = 3 - 3x$ for $0 \leq x \leq 1$.

a) Sketch the curve and use geometry to compute its length.

b) Use integration to compute the length of C .

c) Compute the surface area obtained by rotating the curve C about the x -axis.

d) Compute the surface area obtained by rotating the curve C about the y -axis.

e) The total surface area of a right cone of height h and radius r (not counting the circular base) is given by $S = \pi r \sqrt{r^2 + h^2}$. Derive this formula for the surface area of the cone using calculus.

- f) Determine the radius and height of the cones in parts (c) and (d) and then use the formula you created in part (e) to confirm the calculation of the surface area that you found in parts (c) and (d).

5. For the right values of a and b ($b \geq 0$) the line $y = x + b$ will have an arc length on the interval $[0, a]$ that is equal to the surface area of the solid obtained by rotating $y = x + b$ about the x -axis on this same interval.
- a) Confirm that this is true if $b = 0$ and $a = \frac{1}{\pi}$.

- b) For any given value of a determine the value of b that makes the given scenario true. That is, find a formula for b in terms of a .